

PRATEEK SAHU

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EDUCATION

University of Texas at Austin

M.Sc/Ph.D., Computer Architecture.

Indian Institute of Technology, Kanpur

Bachelor of Engineering, Electrical.

August 2017 - Present

GPA: 3.79/4

July 2011 - May 2015

GPA: 8.2/10

SELECT PUBLICATIONS

Sahu, P.*, Banerjee, S.*, Luo, M., Vahldiek-Oberwagner, A., Yadwadkar, N.J., Tiwari, M. (2024, Nov). *SoK: A systems perspective on compound ai threats and countermeasures* arXiv '24. [pdf]

Sahu, P., Wei, S., Yadwadkar, N.J., Tiwari, M. (2025, Mar). *Understanding Sidecars in Cloud Orchestration* SESAME '25, co-located with EuroSys '25. [pdf]

Behnia, M., **Sahu, P.**, Paccagnella, R., Yu, J., Zhao, Z., Zou, X., Unterluggauer, T., Torrellas, J., Rozas, C., Morrison, A., McKeen, F., Liu, F., Gabor, R., Fletcher, C.W., Basak, A., Alameldeen, A. (2021, April). *Speculative Interference Attacks: Breaking Invisible Speculation Schemes* In Proceedings of the 26th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (pp. 1046-1060). [pdf]

Harris, A., Wei, S., **Sahu, P.**, Kumar, P., Austin, T., Tiwari, M. (2019, October). *Cyclone: Detecting Contention-Based Cache Information Leaks Through Cyclic Interference*. In Proceedings of the 52nd Annual IEEE/ACM International Symposium on Microarchitecture (pp. 57-72). ACM. [pdf]

WORK EXPERIENCE

Interests: Systems Security, Malware Detection, Secure AI

SPARK Reasearch Lab, University of Texas at Austin

Graduate Research Assistant, Advised by Prof. Mohit Tiwari

Spring 2018 - Present

- **Micro-architectural Side Channels**

- *Speculative Interference Attacks* exploits younger instruction affecting older instruction latency.
- Proof-of-concept attacks on caches that undermines defense mechanisms.
- *Cyclone* is a side-channel malware detector that tracks cyclic contention events across security domains.
- Evaluated against cache based side and covert channels on gem5 simulation environment.

- **Ransomware Detection**

- *Lifecycle Aware Ransomware Detector* monitors cross layer system behavior across malware lifecycle.
- Relies on hardware, system and network events to improve detection confidence.
- Is trained on MITRE malware lifecycle stages for early detection and lower data loss.

- **Threat Modeling Generative AI systems**

- Systematize cross layer threat vectors and model privileged adversaries in modern AI inference pipelines.
- Explore system attack vectors to bypass application layer defense mechanisms to introduce prompt injection attacks.

Intel Labs, Intel Corp.

Graduate Technical Intern

June 2020 - Nov 2020

- **Secure Accelerator Design:** Design of secure data offload to Accelerators

- System design for remote attestation
- Proof of concept designs for key provisioning and secure data offload

SELECT COURSE PROJECTS

Verilog system design for 32-bit x86 ISA

Microarchitecture Course, Dr. Y. Patt

Intelligent instruction duplication for side-channel defence [Report]

Security Course, Dr. M. Tiwari

AWARDS AND SERVICE

- Qualcomm Innovation Fellowship recipient for 2022.
- Kishore Vaigyanik Protsahan Yojana fellowship by Department of Science and Technology, India.

★ Serving on the Technical Program Committee for MICRO 2026.